

EECS 841: Computer Vision

Lecture 4

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Outline (Chapter 3)

Lecture 4_ Geometric Transform _Part I

Image Transformation: Different ways to transform an image into another image

- Geometric transform
- Point transform

Geometric Transform

Basic Geometric transform: the pixel intensity values do not change → only the location/index is changed for a particular intensity

Example:

- Flip: upside down (top becomes bottom, bottom becomes top)
- Flop: mirroring (left becomes right, right becomes left)
- +90 degree Rotate: (width becomes height, height becomes width)

Point Transform

Point transform: the pixel intensity does change in output image after the processing

Example:

- Graylevel Transformations
- Graylevel Histograms
- Multispectral Transformations
- Multi-Image Transformations
- Change Detection
- Compositing
- Interpolation

Geometric Transformation

Flipping and Flopping

- The simplest geometric transformation is to reflect the image about a horizontal or vertical axis passing through the center of the image.
 - If the axis is horizontal, the transformation **flips** the image upside down.
 - If the axis is vertical, the transformation **flops** the image to produce a right-to-left mirror image.

$$\begin{bmatrix} 128 & 78 & 174 \\ 181 & 48 & 77 \\ 109 & 49 & 138 \end{bmatrix} \xrightarrow{\text{FLOP}} \begin{bmatrix} 174 & 78 & 128 \\ 77 & 48 & 181 \\ 138 & 49 & 109 \end{bmatrix}$$

↓ FLIP

$$\begin{bmatrix} 109 & 49 & 138 \\ 181 & 48 & 77 \\ 128 & 78 & 174 \end{bmatrix}$$

Geometric Transformation: identify Flip/Flop



Q: Which one is Flip and which one is Flop?

Image

Flip

Flop

Flip

w = no. of columns = 5

h = no. of rows = 4

	j	→				
		0	1	2	3	4
i	0	1	2	3	4	5
	1	6	7	8	9	10
	2	11	12	13	14	15
	3	16	17	18	19	20

$$I(x,y) = I(c,r) = I(j,i)$$

Q: How to develop a formula with which for a given $I(x,y)$, we can find corresponding $I'(x',y')$ if we flip the image?

In flipping, we don't change the values of columns. Only the values in each row change. Thus, if I' is the flipped image and each pixel is $I'(x',y')$ then:

$$I'(x',y') \leftarrow I(x,y)$$

Now, $x = x'$

But, $y' = ?$

How about:

$y' = h - 1 - \text{current row position}$

$$= h - 1 - y$$

(we are doing "-1" because we start the index from 0)

Flip

w = no. of columns = 5

h = no. of rows = 4

	j					
		0	1	2	3	4
i	0	1	2	3	4	5
	1	6	7	8	9	10
	2	11	12	13	14	15
	3	16	17	18	19	20

$$I(x,y) = I(c,r) = I(j,i)$$

Q: How to develop a formula with which for a given $I(x,y)$, we can find corresponding $I'(x',y')$ if we flip the image?

Now, let's try the formula with a particular value

Let's look at the flipped image first. Then we can work on the formula

$$I'(x',y') = I(j,i')$$

w = no. of columns = 5

h = no. of rows = 4

	j					
	0	1	2	3	4	
i	0	16	17	18	19	20
	1	11	12	13	14	15
	2	6	7	8	9	10
	3	1	2	3	4	5

Flip

w = no. of columns = 5

h = no. of rows = 4

	j	0	1	2	3	4
i	0	1	2	3	4	5
1	6	7	8	9	10	
2	11	12	13	14	15	
3	16	17	18	19	20	

$$I(x,y) = I(c,r) = I(j,i)$$

Q: What location should $I(3,2)$ be in $I'(x',y')$?

$$I(3,2) = I(x,y)$$

w = no. of columns = 5

h = no. of rows = 4

	j	0	1	2	3	4
i	0	16	17	18	19	20
1	11	12	13	14	15	
2	6	7	8	9	10	
3	1	2	3	4	5	

The answer should be $I'(3,1)$

Now, let's try the formula with the particular value

Flip

w = no. of columns = 5

h = no. of rows = 4

	j	0	1	2	3	4
i	0	1	2	3	4	5
1	6	7	8	9	10	
2	11	12	13	14	15	
3	16	17	18	19	20	

$$I(x,y) = I(c,r) = I(j,i)$$

Q: What location should $I(3,2)$ be in $I'(x',y')$?

$$I(3,2) = I(x,y)$$

$$\begin{aligned} I'(x',y') &= I'(x, h-1-y) \\ &= I'(3, 4-1-2) \\ &= I'(3, 1) \end{aligned}$$

h = no. of rows = 4

	j	0	1	2	3	4
i	0	16	17	18	19	20
1	11	12	13	14	15	
2	6	7	8	9	10	
3	1	2	3	4	5	

It does work!

Flip

ALGORITHM 3.1 Flip an image by reflecting about a horizontal axis

FLIPIMAGE (I)

Input: image I of size $width \times height$

Output: upside-down image I'

1 $I' \leftarrow \text{ALLOCATEIMAGE}(width, height)$

➤ Allocate memory for output image.

2 **for** $(x, y) \in I$ **do**

➤ For each pixel in input image,

3 $I'(x, height - 1 - y) \leftarrow I(x, y)$

set corresponding pixel in output image.

4 **return** I'

➤ Return output image.

- Upside down
- You are modifying the row position of a pixel in I to another row in I'
- The column remains same

Flop

ALGORITHM 3.2 Flop an image by reflecting about a vertical axis

FLOPIMAGE (I)

Input: image I of size $width \times height$

Output: mirror-reversed image I'

1 $I' \leftarrow \text{ALLOCATEIMAGE}(width, height)$

➤ Allocate memory for output image.

2 **for** $(x, y) \in I$ **do**

➤ For each pixel in input image,

3 $I'(width - 1 - x, y) \leftarrow I(x, y)$

set corresponding pixel in output image.

4 **return** I'

➤ Return output image.

- Mirroring
- Right becomes left & Left becomes right
- You are modifying the column position of a pixel in I to another column in I'
- The row remains same

Flip-flop?

(upside down and then mirroring)

Figure 3.1 An image, and the result of flipping, flopping, and flip-flopping.



Image



Flip-flop

Stan Birchfield



Image



Flip-flop

Stan Birchfield

Rotate (90 degrees Clockwise): Try!

- The transformations are expressed as functions that define the mapping between each input pixel $I(x, y)$ and its corresponding output pixel $I'(x', y')$:

$$I'(\text{height} - 1 - y, x) = I(x, y)$$



0°



+90°

Rotate: 90 degrees Clockwise

h = no. of rows = 4

w = no. of columns = 5

	j	0	1	2	3	4
i	0	1	2	3	4	5
1	6	7	8	9	10	
2	11	12	13	14	15	
3	16	17	18	19	20	

$$I(x,y) = I(c,r) = I(j,i)$$

Q: What location should $I(4,1)$ be in $I'(x',y')$?

$$I(4,1) = I(x,y)$$

Thus,

$$x = 4$$

$$y = 1$$

no. of rows = 5

no. of columns = 4

	j	0	1	2	3
i	0	16	11	6	1
1	17	12	7	2	
2	18	13	8	3	
3	19	14	9	4	
4	20	15	10	5	

Ans should be (2,4)

Let's see if the formula works!

$$\begin{aligned} I'(x',y') &= I'(h-1-y, x) \leftarrow I(x,y) \\ &= I'(4-1-1, 4) \\ &= I'(2, 4) \end{aligned}$$

It does work!

Downsampling and Upsampling

Downsampling and Upsampling

Purpose?

- **Downsample** an image to produce a smaller image than the original:
- **Upsample** an image to produce a larger image than the original:

Downsampling: compression

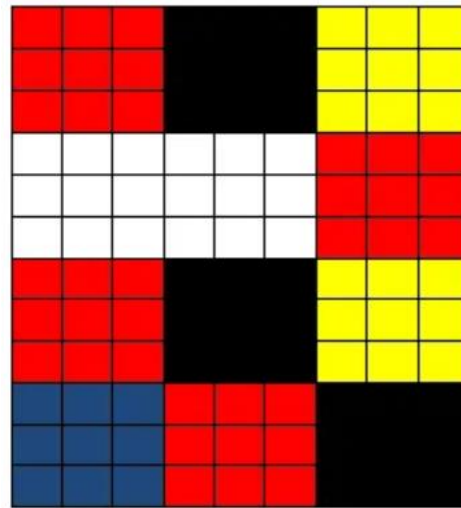
Upsampling: elaboration

Figure 3.6 LEFT: An image and the result of downsampling by a factor of 2 and 4, respectively, in each direction. RIGHT: A cropped region and the result of upsampling by a factor of 2 and 4, respectively, in each direction.



Factor needed for sampling

Question:



How to determine the level of compression and elaboration?

A: Scaling factor!

Width x 3
Height x 3

Which one is upsampling?

Ans: 

Which one is downsampling?

Ans: 

Point Transform: Next class

Point transform: the pixel intensity does change in output image after the processing

Example:

- Graylevel Transformations
- Graylevel Histograms
- Multispectral Transformations
- Multi-Image Transformations
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- Interpolation